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Title: The Effect of Loneliness on Distinct Health Outcomes: A Comprehensive Review and Meta-Analysis.

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Abstract: Importance: Loneliness has been associated with adverse health outcomes. However, few studies have evaluated its comparative effects on distinct health outcomes.

Objective: The primary objective was to evaluate the comparative effects of loneliness on multiple health outcomes. The literature was qualitatively reviewed to identify risk factors associated with loneliness, explore mechanisms, and discuss potential evidence-based interventions for targeting loneliness.

Data Sources: A systematic review and meta-analysis of studies were conducted to examine the association between loneliness and health outcomes. We searched the databases Ovid MEDLINE/PubMed, Embase, and PsycINFO from inception to April/01/2019 for relevant peer-reviewed publications.

Study Selection: We selected studies that measured the association between subjective ratings of loneliness (i.e., UCLA Loneliness Scale or de Jong Gierveld Loneliness Scale) and one or more health outcome(s). Only validated measurement scales for health outcomes were used.

Data Extraction and Synthesis: Data was extracted by four independent reviewers using a standardized data extraction form. A total of 114 identified studies were systematically evaluated to extract study characteristics, risk of bias, and loneliness-health associations required for the calculation of pooled effects (i.e., correlations).

Main Outcome(s) and Measures: Loneliness and health outcomes were measured. Health outcomes were broadly defined to include measures of mental health (i.e., depression, anxiety, suicidality, general mental health), general health (i.e., overall self-rated health), well-being (i.e., quality of life, life satisfaction), physical health (i.e., functional disability), sleep, and cognition.

Results: Loneliness had medium to large effects on all health outcomes, with the largest effects on mental health and overall well-being. No significant effects of age, health condition, or stigma were observed. However, we observed a significant effect of gender on the association between loneliness and cognition (i.e., more pronounced in studies with a

greater proportion of males). We also observed a significant effect of geographical distribution on the associations between loneliness and overall well-being, sleep, and cognition (i.e., lower among European nations and higher among Asian nations).

Conclusions and Relevance: Loneliness is associated with adverse effects on health, particularly mental health outcomes. Future research should explore the potential moderating effects of gender and geographical distribution. From a public health perspective, the adequate training of health care providers (HCPs) to perceive, detect, and respond to loneliness among patients should be prioritized.

The Effect of Loneliness on Distinct Health Outcomes: A Comprehensive Review and Meta-Analysis

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Key Points:

Question: What are the comparative associations between loneliness and various health outcomes (i.e., mental health, physical health, general health, well-being, sleep, and cognition)?

Findings: A comprehensive meta-analysis revealed medium to large effects of loneliness on all health outcomes, with the largest effects on mental health outcomes and overall well-being.

Meaning: There are strong associations between loneliness and adverse health outcomes; healthcare physicians should be adequately trained to perceive and respond to loneliness.

Abstract

Importance: Loneliness has been associated with adverse health outcomes. However, few studies have evaluated its comparative effects on distinct health outcomes.

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Results: Loneliness had medium to large effects on all health outcomes, with the largest effects on mental health and overall well-being. No significant effects of age, health condition, or stigma were observed. However, we observed a significant effect of gender on the association between loneliness and cognition (i.e., more pronounced in studies with a greater proportion of males). We also observed a significant effect of geographical distribution on the associations between loneliness and overall well-being, sleep, and cognition (i.e., lower among European nations and higher among Asian nations).

Conclusions and Relevance: Loneliness is associated with adverse effects on health, particularly mental health outcomes. Future research should explore the potential moderating effects of gender and geographical distribution. From a public health perspective, the adequate training of health care providers (HCPs) to perceive, detect, and respond to loneliness among patients should be prioritized.

1. Introduction

Loneliness has recently been reported to affect as much as one third of individuals in industrialized countries ¹, and has been commonly reported among youth and older adults. ² Loneliness has been negatively associated with multiple health outcomes, including mental and physical health ³, sleep ⁴, cognition ⁵, and cardiovascular health ⁶. In addition, meta-analyses have reported that lonely individuals exhibit higher rates of morbidity and mortality ^{6,7}, including poorer quality of life and functional decline ⁸. Loneliness has also been associated with cognitive impairment and increased prevalence of mental health disorders, such as depression ⁹ and anxiety ¹⁰.

Although loneliness lacks a consensual definition, most researchers agree that loneliness can be described as the subjective discrepancy between an individual's existing versus desired social relationships ⁸. It is important to note, however, the distinction between loneliness and social isolation. While loneliness refers to a *subjective* feeling, social isolation is an *objective* measure of social connectedness; they are distinct constructs that cannot be used interchangeably. For example, it is possible for those who are more objectively isolated (e.g., live alone, have smaller social networks) to not feel lonely, while those who are less objectively isolated (e.g., live with others, have larger social networks) feel lonely. In this sense, loneliness can be described as perceived social isolation⁸.

While the association between loneliness and adverse health outcomes has been well researched, literature examining the differential effects of loneliness on distinct health outcomes (e.g., mental health versus physical health outcomes) is less extensive. Given the high prevalence of loneliness, it is important for researchers to thoroughly investigate the comparative associations between loneliness and adverse health outcomes. A comprehensive understanding of

these associations may inform clinicians with respect to strategically screening and targeting loneliness in clinical settings, and is relevant insofar as loneliness is not currently evaluated and treated in routine clinical care.

The primary objective of the current systematic review and meta-analysis was to compare and evaluate the association between loneliness on various health outcomes. The literature was qualitatively reviewed to address the following secondary objectives: identify risk and protective factors associated with loneliness; explore mechanisms mediating the association between loneliness and adverse health outcomes; and discuss potential evidence-based interventions for targeting loneliness.

2. Methods

The current systematic review and meta-analysis adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement for transparent and comprehensive reporting of study methods and results¹¹. We evaluated studies that measured the association between loneliness and one or more health outcome(s). Using a random effects model, we calculated a pooled estimate of the effect of loneliness on the following health outcomes: depression, anxiety, suicidality, general mental health, general health, well-being, physical health, sleep, and cognition. Only studies using validated measures were included. For each health outcome, we also examined factors that may have moderated the impact of loneliness via subgroup and meta-regression analyses.

2.1 Search Strategy

We searched the databases Ovid MEDLINE/PubMed, Embase, and PsycINFO from inception to April/01/2019 for relevant peer-reviewed publications. Titles and abstracts were

searched using the following medical search heading terms: “Loneliness” AND “Health”.

Reference lists of identified articles and relevant review articles were manually searched for additional articles.

2.2 Eligibility Criteria

The titles and abstracts of all identified publications were screened for eligibility. We aimed to identify studies that measured the association between loneliness and one or more health outcome(s). Health outcomes were broadly defined to include measures of mental health (depression, anxiety, suicidality, general mental health), general health (overall self-rated health), well-being (quality of life, life satisfaction), physical health (functional disability), sleep, and cognition. Mortality was excluded as several recent meta-analyses have been conducted on the association between loneliness and mortality^{6,7}. Both objective and subjective measures of health were included. Herein, we limited our focus to studies evaluating loneliness, and excluded studies evaluating related constructs (e.g., social isolation, social support). Additionally, we only included studies that evaluated outcomes derived from clinical interviews and/or valid self-report (e.g., depression severity, suicidality). We purposefully did not include studies evaluating physiological and/or biological markers (e.g., changes in blood pressure or biomarkers) as outcomes; however, blood biomarkers (e.g., cortisol, inflammatory molecules) are discussed in the context of mechanisms mediating the association between loneliness and health outcomes. Detailed inclusion/exclusion criteria are reported in **Table 1**. The full-texts of all identified articles were reviewed for eligibility. Studies that did not report, evaluate, or sufficiently describe the foregoing information were excluded from analysis.

2.3 Data Extraction

Correlation values between loneliness and the health outcome(s) were extracted from each study. Using a standardized data extraction form, data was extracted from included studies by four independent reviewers to systematically evaluate study characteristics, risk of bias, and loneliness-health associations required for the calculation of pooled effects (i.e., correlations). The following information was collected: study information (author, population, sample size, age of participants, study design, and study quality); loneliness scale title; health outcome scale title; and relevant statistics. Where studies reported an unadjusted odds ratio instead of a correlation value, the corresponding correlation was calculated. Where studies reported more than one health outcome, the correlation between loneliness and each individual health outcome was considered separately. Where studies measured loneliness and health outcome(s) at multiple time points (i.e., longitudinal studies), the correlation was reported for all cross-sectional correlations (e.g., loneliness T1 x health outcome T1; loneliness T2 x health outcome T2), and for all longitudinal correlations wherein loneliness was measured prior to the health outcome (e.g., loneliness T1 x health outcome T2). Where there was disagreement on study quality assessment, consensus was reached through discussion.

2.4 Statistical Analyses

All analyses were conducted using *Comprehensive Meta-Analysis 3.0*. A random effects model was used given the expected variability among included studies. A separate pooled analysis was conducted for each health outcome category. Pooled correlations (r) were calculated to estimate the effect size. The mental health category was divided into subgroups to delineate the effects of loneliness on various mental health outcomes (i.e., depression, anxiety, suicidality, and general mental health).

Subgroup analyses for categorical variables (i.e., age category, geographical distribution, health condition, and belonging to a stigmatized group) and meta-regression analyses for continuous variables (i.e., mean age, gender distribution) were conducted to examine for their potential moderating effects. Critical values for pooled effect sizes were set at 0.05. Homogeneity in effect sizes was tested using Cochran's Q statistic and heterogeneity was quantified using the I^2 statistic, where 25%=small, 50%=moderate and 75%=high heterogeneity.

2.4 Quality Assessment

Risk of bias was assessed using the *Newcastle-Ottawa Quality Assessment Scale*. Higher study quality is indicated by the number of stars awarded¹². The results of the quality assessment are displayed in **eTable 1**.

Results

Search Results

The initial search yielded a total of 3016 unique records. After title and abstract review, 2585 records were excluded, and 431 full-text publications were assessed for eligibility. After full-text review, 317 records were excluded. Additional studies were identified via manual search of reference lists. A total of 114 studies evaluating the effects of loneliness on various health outcomes were included in the current meta-analysis (**Figure 1**).

Details of included studies

Details of each study are displayed in **eTable 1** ($k=114$, pooled $N=18,512$). The majority of studies were cross-sectional ($k_{\text{cross-sectional}}=105$; $k_{\text{longitudinal}}=9$), and included older adults (≥ 50

years). Out of 114 studies, only 21 were conducted in youth (<25 years). Mental health outcomes, particularly depression, were most commonly studied in relation to loneliness. All studies were conducted in North America, Europe, Asia, and/or Oceania. Sample characteristics varied across studies; associations were reported in a variety of different groups including, but not limited to, community-dwelling seniors, stigmatized groups, patients with chronic conditions (e.g., HIV, cancer, Alzheimer's disease, severe mental illness), and university/college students.

Main analyses: Association between loneliness and health outcomes

Figure 2 displays the results of the main analyses; forest plots and pooled correlations for the association between loneliness and depression(**2a**), anxiety(**2b**), suicidality(**2c**), general mental health(**2d**), general health(**2e**), well-being(**2f**), physical health(**2g**), sleep(**2h**), and cognition(**2i**).

The random effects meta-analyses revealed medium to large effects of loneliness on all health outcomes, with the largest effects on mental health and overall well-being, and smaller effects on sleep and cognition. **Table 2** compares the pooled correlation values for each respective health outcome.

As shown in **Figure 2a**, the pooled correlation between loneliness and depression was -0.497, with a 95% confidence interval (CI) of -0.524 to -0.469. Similarly, for anxiety(**2b**), suicidality(**2c**), and general mental health(**2d**), the pooled correlations were -0.417 (95%CI: -0.469 to -0.362), -0.516 (95%CI: -0.629, -0.381), and -0.489 (95%CI: -0.556, -0.415), respectively. The pooled effect of loneliness on general health(**2e**) and overall well-being(**2f**) were also large [-0.358 (95%CI: -0.470, -0.286), and -0.476 (95%CI: -0.517, -0.433), respectively]. In contrast, the pooled effect of loneliness on physical health(**2g**), sleep(**2h**), and

cognition(2i) were relatively lower [-0.301 (95%CI: -0.348, -0.253), -0.293 (95%CI: -0.335, -0.250), and -0.189 (95%CI: -0.262, -0.114), respectively].

There was high heterogeneity in data across studies for all health outcome categories ($I^2 > 75\%$) except sleep, which had moderate heterogeneity ($I^2 = 54.08\%$) (**Table 2**). There was also evidence of possible publication bias for the following categories: depression, suicidality, general mental health, physical health, physical health, and well-being, as indicated by Egger's regression tests ($p < 0.05$) (**Table 2**). For all health outcomes, funnel plots appeared relatively symmetrical and showed a higher proportion of studies with high precision (i.e., low standard error) (**eFigure 1, a-i**).

Subgroup and meta-regression analyses

Interestingly, age (i.e., age category and mean age) did not significantly moderate did not significantly moderate the associations between loneliness and any of the measured health outcomes (all $ps > 0.05$). We also found that having a chronic health condition (e.g., HIV, cancer, Alzheimer's disease, severe mental illness) or being a member of a stigmatized group did not significantly influence the association between loneliness and health, for any of the measured health outcomes.

We found that loneliness had a greater negative impact on cognition in studies with a larger distribution of males, suggesting that the association between loneliness and cognition may be moderated by gender. However, no effect of gender was observed for other health outcomes.

We also observed an effect of geographical distribution. We found that the following associations differed significantly based on geographical distribution: loneliness and overall

well-being ($p=0.004$), loneliness and sleep ($p=0.006$), and loneliness and cognition ($p<0.001$).

Tables 3a and **3b** display the results of the subgroup and meta-regression analyses. Across these associations, we found that the negative impact of loneliness on health was lower in Europe and higher in Asia.

Discussion

Summary of results

Overall, the main analyses revealed medium to large effects of loneliness on all health outcomes, with the largest effects on mental health and overall well-being. However, there was high heterogeneity among included studies, at least some of which can be attributed to differences in sample characteristics.

Sub-group and meta-regression analyses revealed a significant effect of gender on the association between loneliness and cognition (i.e., the negative impact of loneliness was greater in studies with a greater proportion of males); however, no significant effects of age, health condition, or stigma were observed. The latter non-significant findings may be a consequence of several assumptions that we made regarding the impact of having a chronic health condition, and the particular groups that experience stigma. We categorized individuals who were homeless, mentally ill, affected by a sexually transmitted infection, and/or part of the LGBTQ community as stigmatized, but did not include seniors, despite the fact that many seniors often feel stigmatized. Additionally, we assessed the presence of having a chronic health condition, but were not able to evaluate the *impact* (i.e., stress caused by) having these conditions. In future studies, the moderating effects of these variables should be formally evaluated by implementing

validated scales that specifically measure health- or disability-related stress and/or perceived stigma.

We also observed a significant effect of geographical distribution (i.e., the negative impact of loneliness on health outcomes was lower among European nations and higher among Asian nations). The foregoing observation may be explained by sociocultural differences. For example, Europe is predominantly individualistic (i.e., value independence and autonomy) whereas Asia is predominantly collectivistic (value interdependence and relationships with others, particularly family). To support this, studies have reported higher levels of loneliness in collectivistic compared to individualistic societies¹³, and have found that the factors predicting loneliness differ between collectivistic and individualistic societies. In individualistic societies, loneliness has been associated with a lack of friends and/or the absence of a confidant; however, in collectivistic societies, loneliness has been primarily associated with poor family relationships¹³.

For the majority of health outcomes, there was no data for Oceania; where data was available, there were too few studies to provide sufficient information for comparison. Due to a limited number of studies evaluating the association between loneliness and suicidality, subgroup and meta-regression analyses for suicidality were not conducted.

Potential mechanisms mediating the association between loneliness and health outcomes

Various mechanisms have been suggested to mediate the association between loneliness and adverse health outcomes. Based on our review, however, we found a common unifying

theme of stress; individuals reporting the highest levels of loneliness were often those experiencing significant stress (i.e., sexual minority groups, immigrants, refugees, homeless individuals, individuals with sensory impairments, and individuals with chronic health conditions)^{14–16}. As such, we hypothesize that chronic loneliness leads to chronic stress that triggers a number of downstream inflammatory pathways and poor health behaviours, which eventually culminates in adverse health outcomes. It should be noted, however, that this pathway may operate in reverse (i.e., chronic stress may lead to higher levels of loneliness, which can feedback to influence health outcomes) (**Figure 3**).

Evolutionarily, companionship and social connectedness have been necessary for species survival and self-preservation. As a result, humans evolved a fundamental need to belong and form close bonds with others¹⁷. When we perceive ourselves to be socially isolated, we experience a biological stress response that drives us towards behaviours that promote social inclusion. However, individuals may also experience a maladaptive version of this stress response when there are no objective indicators of social isolation. This is consistent with evidence suggesting that maladaptive cognitive schemas such as hypervigilance (i.e., enhanced sensory awareness and attention to social stimuli)¹⁸ and low self-efficacy (i.e., low confidence in one's ability to exert control over one's situation)^{19,20} may contribute to feelings of loneliness via dysfunctional cognitive processes. Hypervigilance may lead to individuals perceiving the world as more socially threatening, and low self-efficacy may lead to individuals believing that they are incapable of changing their situation (i.e., external locus of control)¹⁸. Research has suggested that these dysfunctional cognitions may be the result of dysregulated prefrontal cortex activity, leading to a loss of cognitive control and inhibitory processes²¹. Ultimately, these

dysfunctional cognitive processes elicit feelings of loneliness and/or negative behaviours such as social withdrawal that subsequently reinforce feelings of loneliness and stress (**Figure 3**).

Chronic stress has been evidenced to impact the endocrine and immune systems, increasing the risk of adverse health outcomes²². Several studies have shown that chronically stressed individuals exhibit altered hypothalamic-pituitary-adrenal (HPA) axis functioning, increased levels of pro-inflammatory markers, and poor health behaviours such as smoking²³, substance use²⁴, sedentary behaviour²⁵, and poor diet²⁶. The pathway from chronic stress to altered HPA axis functioning and subsequent inflammation has been previously described²⁷ and is outlined and explained in **Figure 3**.

Evidence-based interventions for loneliness

Given the adverse health effects associated with chronic loneliness, it may be advantageous to incorporate the assessment and treatment of loneliness into routine clinical care. While a variety of different approaches to treating loneliness have been explored, most research has focused on cognitive-behavioural approaches (e.g., mindfulness-based cognitive therapy [MCBT]) and social interventions.

As mentioned previously, loneliness may result from dysfunctional cognitions. Consequently, researchers have suggested that loneliness may be amenable to cognitive-behavioural approaches, such as MBCT and mindfulness²⁸. For example, mindfulness-based therapies have been shown to be effective at reducing stress and improving social well-being²⁹. In accordance with our conceptual mechanism, other studies have demonstrated that mindfulness-based approaches concurrently reduce levels of stress and circulating proinflammatory markers^{29,30}.

Social interventions aimed at increasing social support and/or promoting social engagement have also been evidenced to reduce feelings of loneliness. For example, social support groups and “social prescribing” exercise programs have been shown to reduce feelings of loneliness by promoting a sense of belongingness and social connectedness.^{31–35, 36} Importantly, it is the social component of these programs, rather than the physical component, that appears to underlie their therapeutic efficacy. For example, Kanamori et al. demonstrated that group exercise improved health outcomes, regardless of the frequency of exercise³⁷.

When designing social interventions that target loneliness, it is important to consider the unique experiences of loneliness across different groups (e.g., loneliness among cancer patients, adolescents, and seniors; intercultural differences in the experience of loneliness), as well as the unique challenges and/or barriers to social engagement experienced by each group. For example, some individuals may lack access to a supportive community and/or have limited social opportunities, while others may be hindered by mobility restrictions or feelings of social anxiety. Assessing the quality of social relationships may also be important, given that many individuals feel lonely despite having a stable social network. Indeed, studies have suggested that the quality of social connections may be relatively more important than the quantity when it comes to perceptions of loneliness³⁸.

With the exception of allopregnanolone (ALLO), few studies examining pharmacological interventions for loneliness have been conducted. ALLO, an endogenous progesterone-derived neurosteroid, is a positive GABA_A modulator and has been shown to have anxiolytic, anesthetic and anticonvulsant effects. Notably, the ALLO mimetic brexanolone was recently FDA-approved for postpartum depression³⁹. Research has demonstrated that ALLO increases during acute stress to negatively regulate the HPA axis; however, during chronic stress, endogenous

production of ALLO decreases, leading to decreased negative regulation and promoting HPA axis hyperactivity^{40–42}. Notably, studies have shown that exogenous administration of ALLO normalizes dysregulated HPA axis functioning⁴³. Taken together, these findings suggest that exogenous ALLO administration may potentially provide therapeutic benefit via negative regulation of the HPA axis in chronically lonely individuals²⁷.

Limitations

The present findings should be considered in the context of the following limitations. First, our study reports only correlation values; therefore, we are unable to establish a causal relationship between loneliness and adverse health outcomes. It is likely that the effects of loneliness on health are complex and multidirectional, with both loneliness contributing to poor health and poor health contributing to increased loneliness.

Moreover, interpretation of the literature is limited by variable definitions of loneliness across studies. For example, studies have defined distinct subtypes of loneliness, such as emotional loneliness (i.e., characterized by the absence of close intimate relationships) and social loneliness (i.e., characterized by the absence of a stable social network)⁴⁴. Numerous other subtypes have also been defined, such as situational, developmental, and internal loneliness⁴⁵. Our study broadly defined loneliness as perceived social isolation, but did not evaluate the effects of loneliness on health based on these different subtypes; it is possible that different loneliness subtypes may have different impacts on health outcomes.

Furthermore, we only included studies that used the UCLA or DJGLS, both of which are self-report scales vulnerable to the influence of personality and psychological factors. In addition, these scales do not report on the chronicity of loneliness, which is a critical aspect

influencing its effect on health outcomes. Researchers should consider developing better scales that capture these critical features.

Conclusions

From a public health perspective, the adequate training of health care providers to perceive, detect, and respond to loneliness among patients should be prioritized. Currently, loneliness is not systematically assessed and treated by clinicians; however, due to its potential adverse effects on health outcomes, the assessment and treatment of loneliness should be integrated into routine clinical care. This may require a shift in public perceptions of loneliness. For example, many patients do not conceptualize loneliness as a health issue to be discussed with their physicians. Likewise, physicians often report difficulties addressing loneliness due to uncertainty related to its clinical assessment and a lack of evidence-based therapeutic options. Future research should better characterize the risk and protective factors associated with loneliness, as well as identifying effective therapeutic options.

Conflict of Interest Disclosures

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All other authors have no conflicts of interest to disclose.

Author Statement

Contributors

Caroline Park and Amna Majeed were primarily involved in the writing, analysis and synthesis of this review. Hartej Gill and Jocelyn Tamura contributed significantly to the writing and synthesis of this review. Jingyi Lu assisted with data extraction. All other authors contributed significantly to the discussion and revision of this work. All authors have read and approved the final version of this review.

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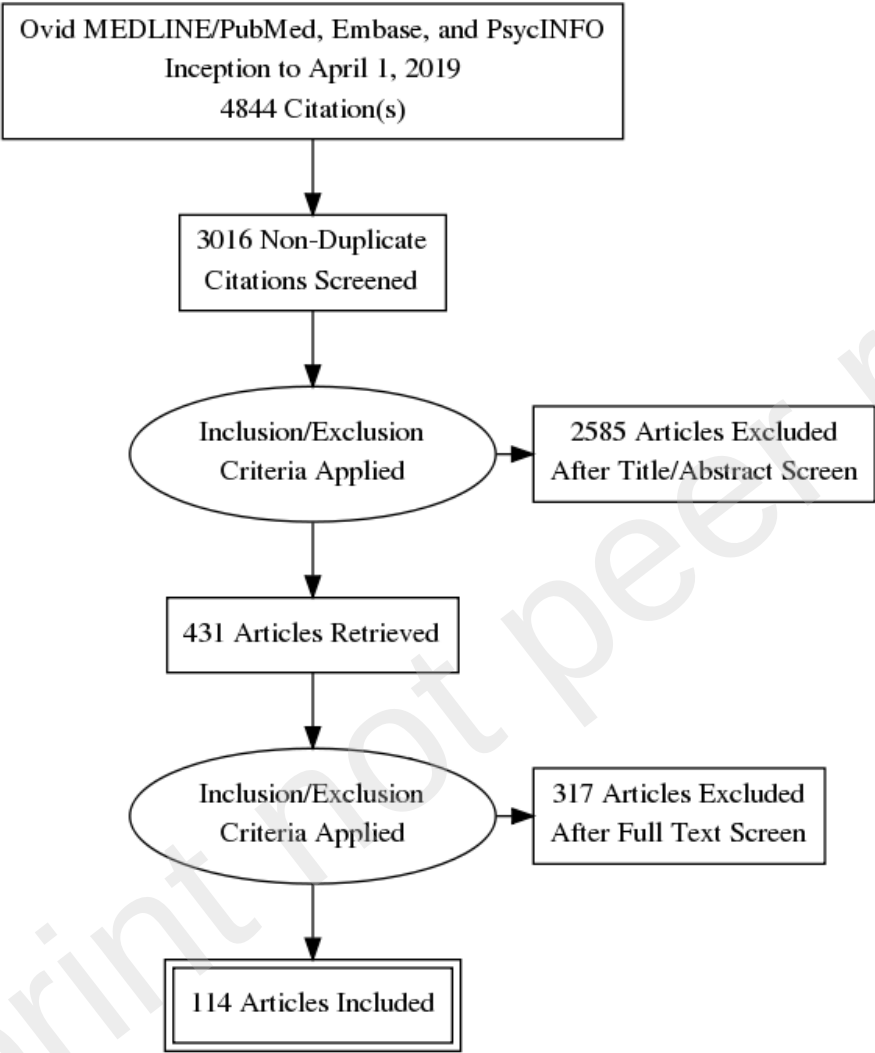
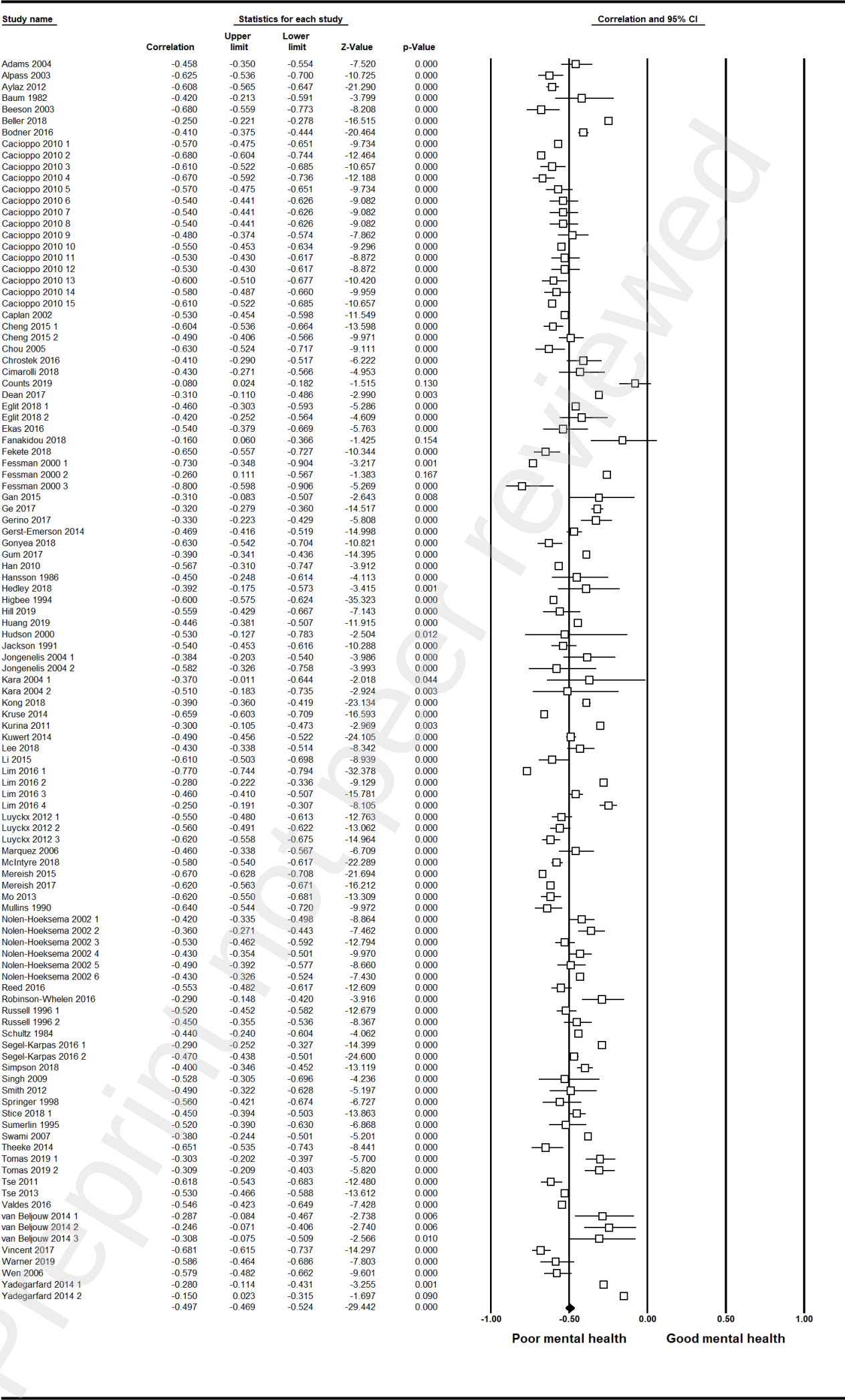
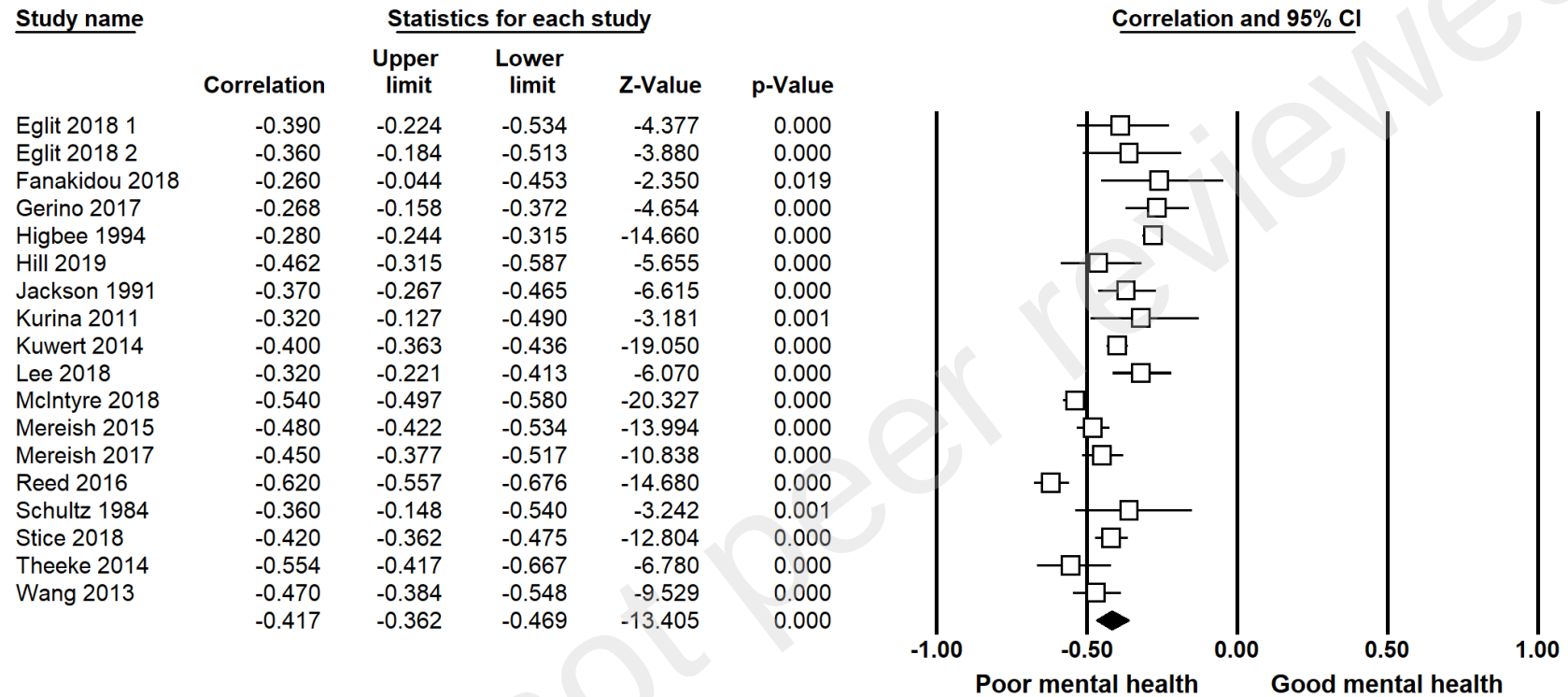


Figure 1. PRISMA flow diagram for study identification and selection.

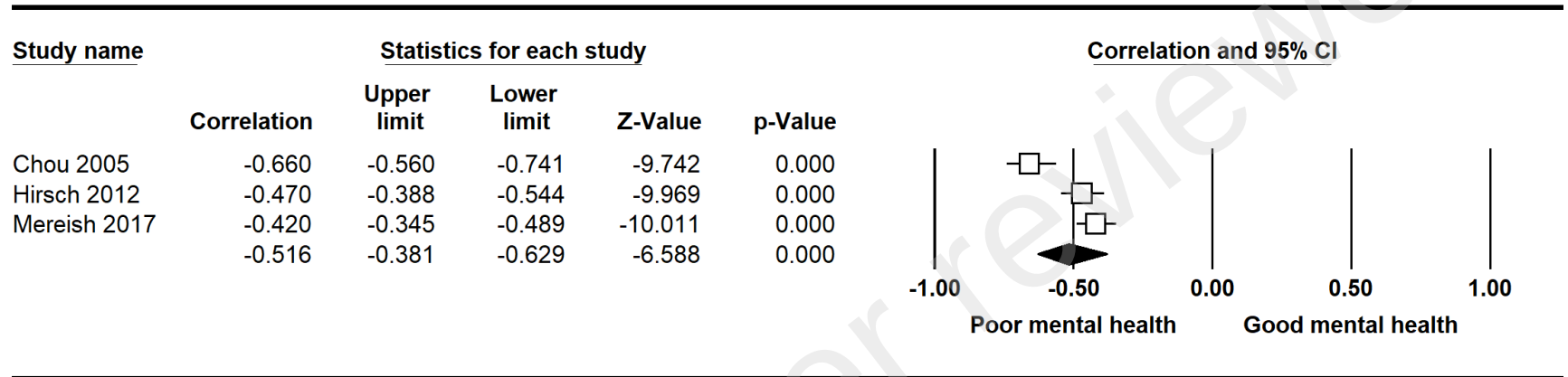
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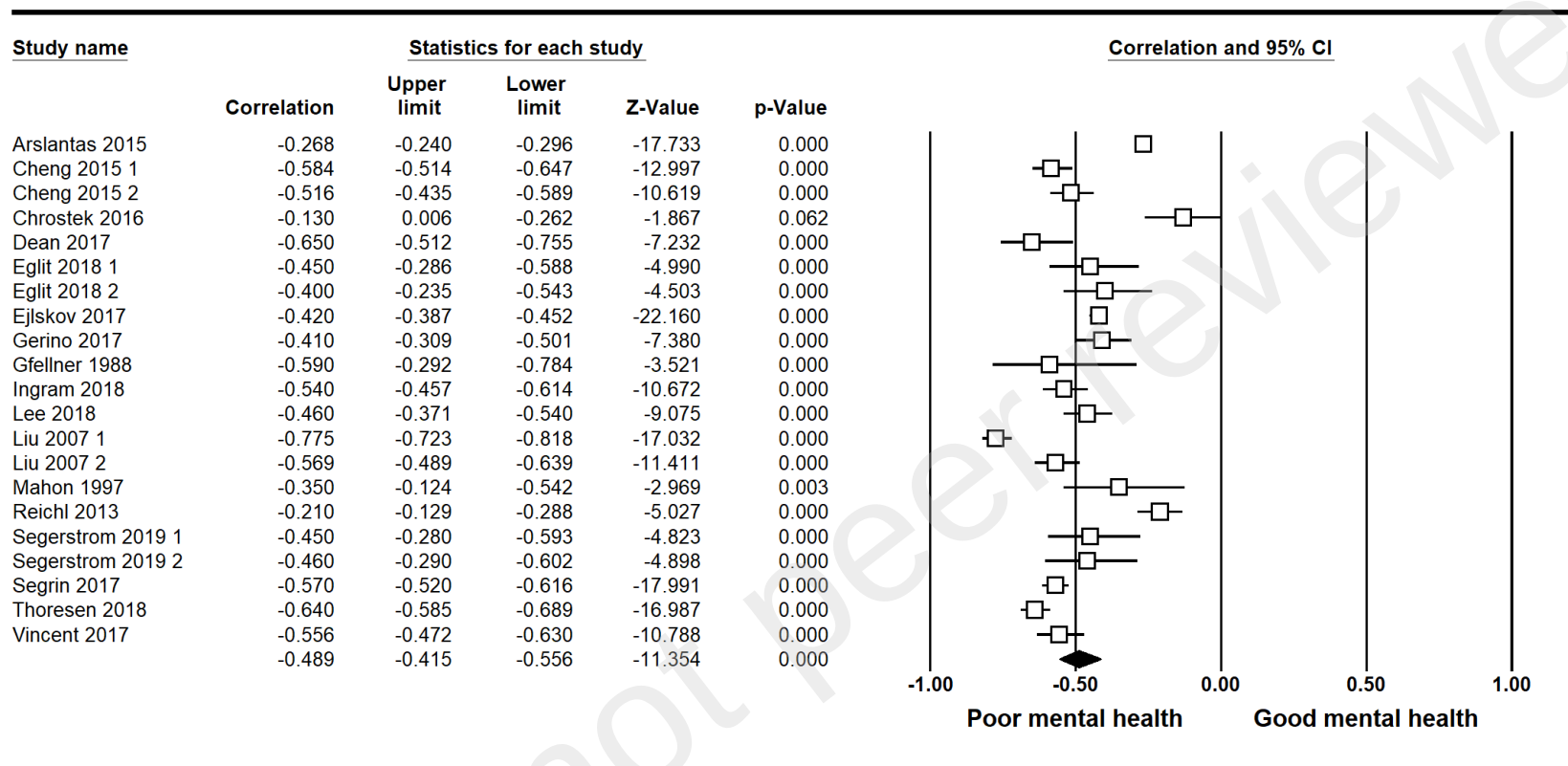
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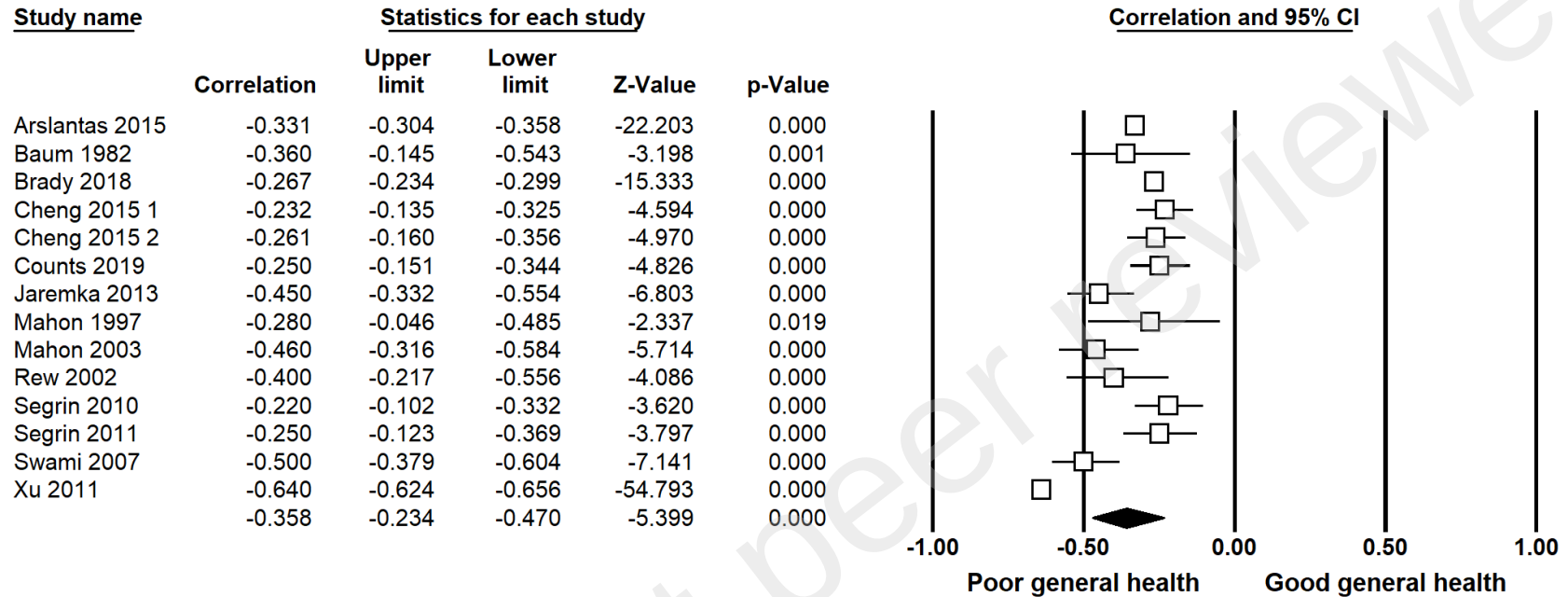
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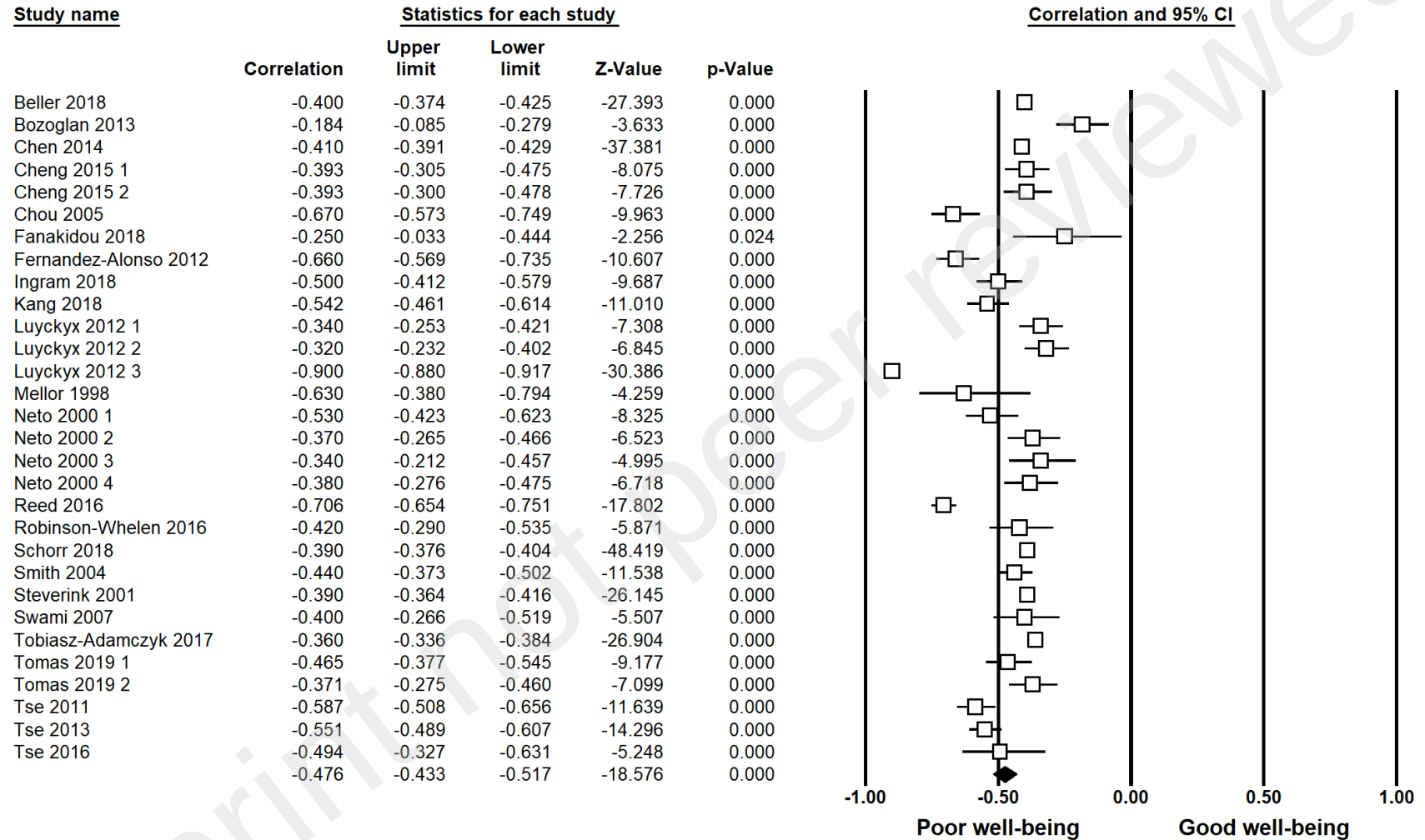
d

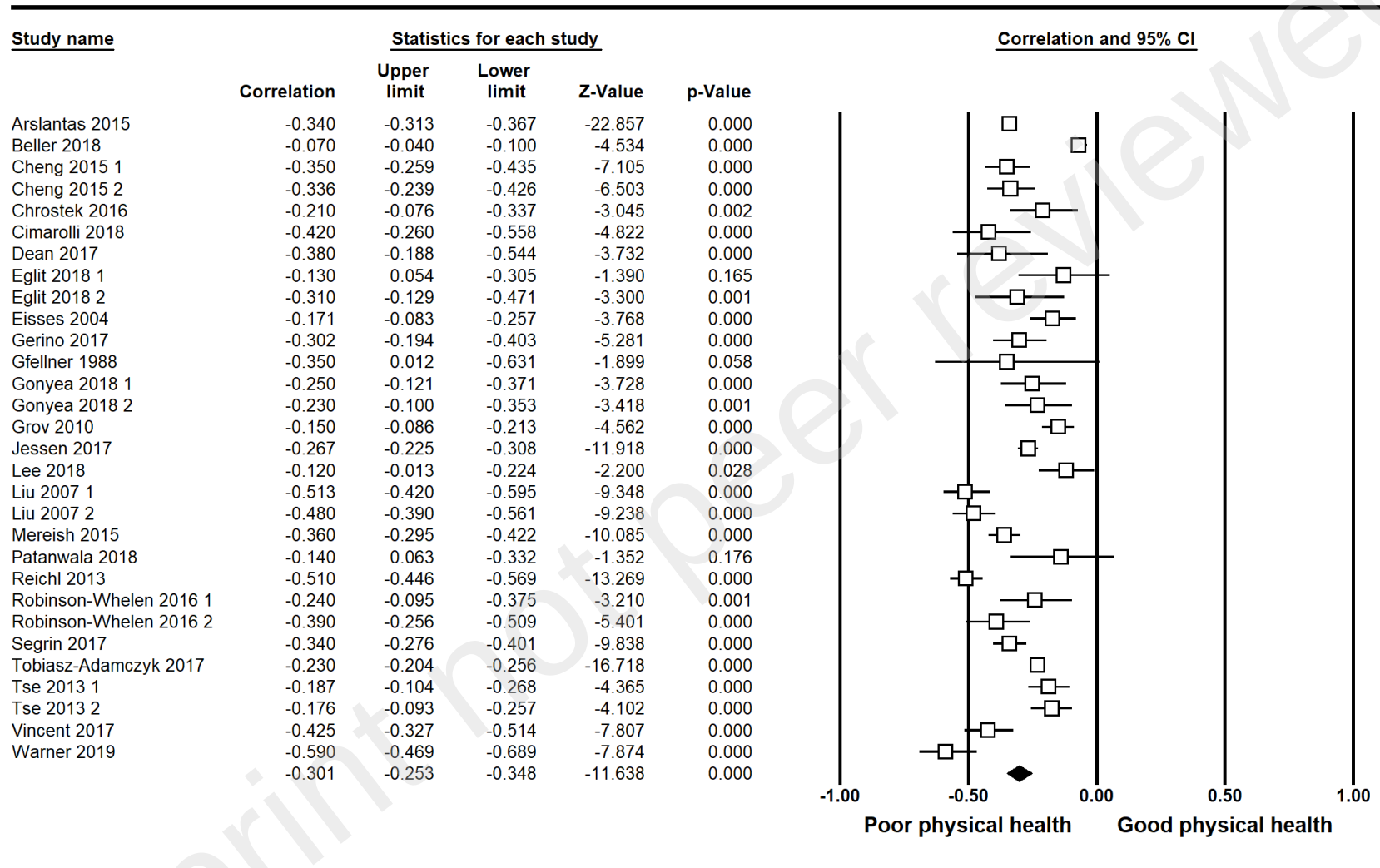


e

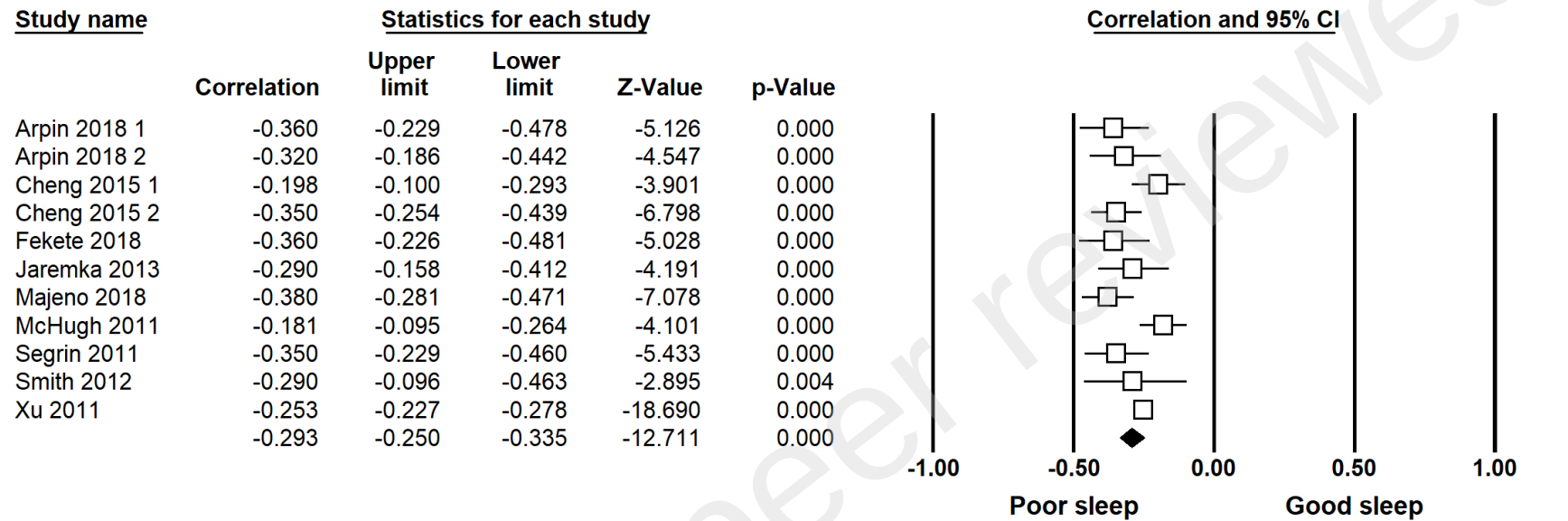


f





h



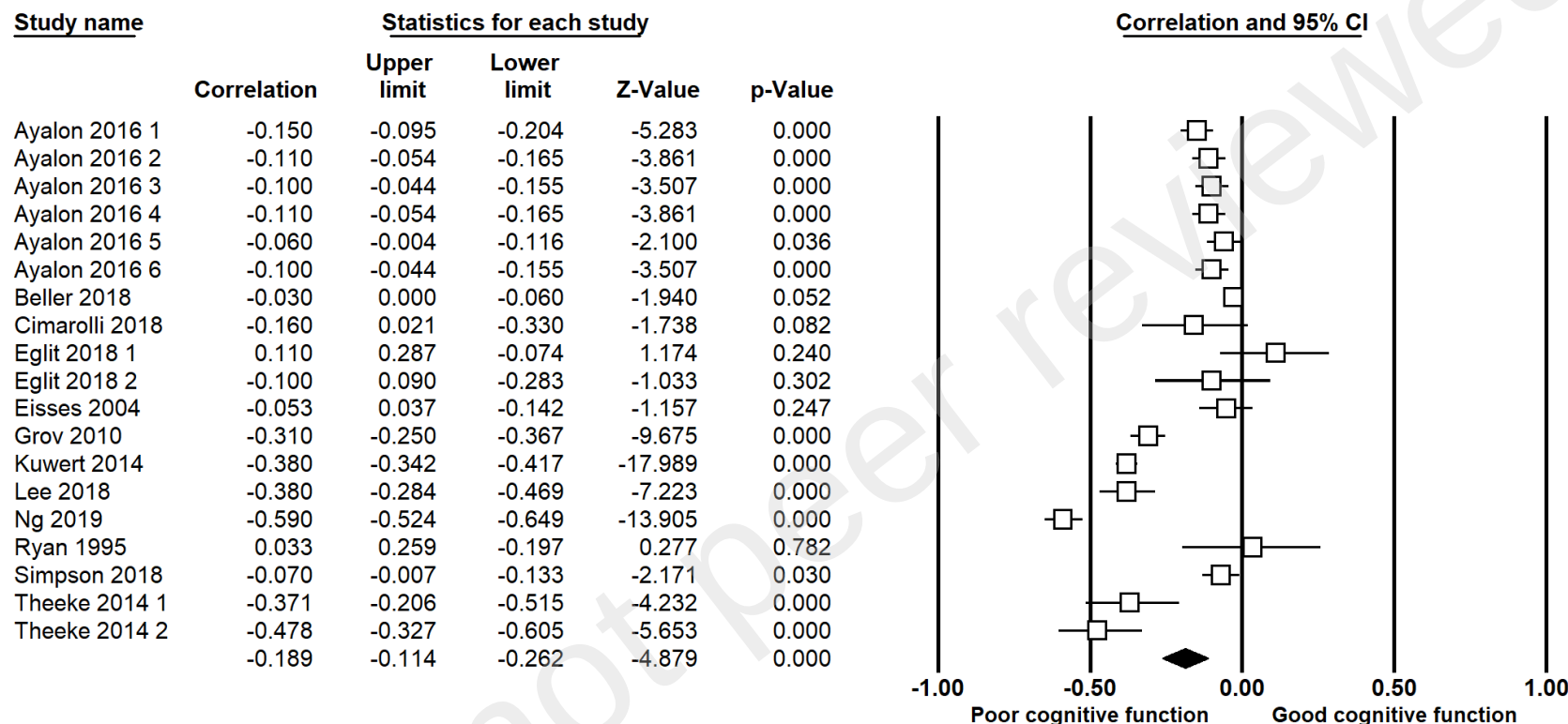


Figure 2. Meta-analysis of the effects of loneliness on disparate health outcomes (i.e., depression (a), anxiety (b), suicidality (c), general mental health (d), general health (e), well-being (f), physical health (g), sleep (h), and cognition (i)). Filled diamond represents the overall pooled effect (i.e., correlation) and 95% CI.

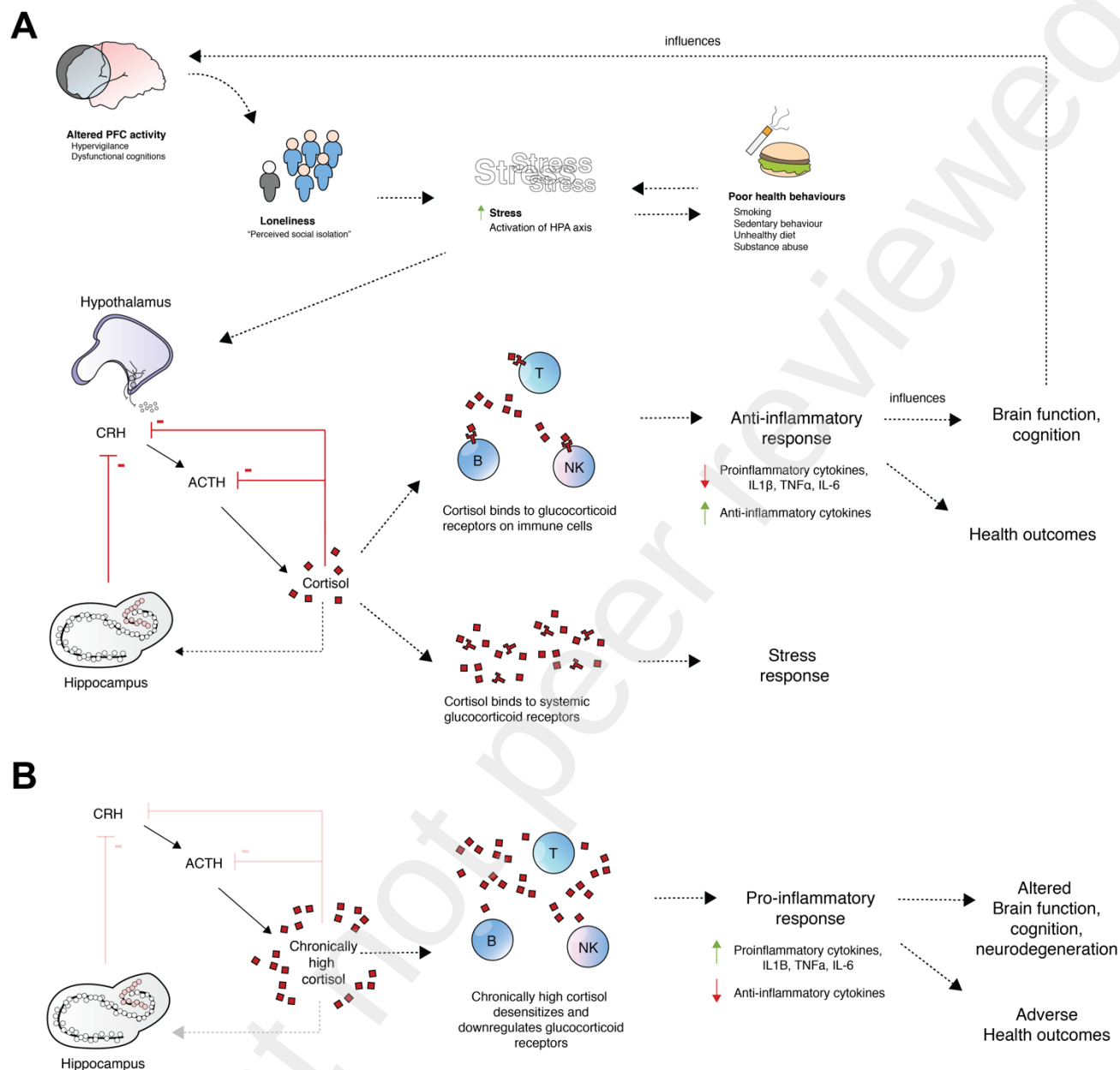


Figure 3. Mechanistic pathway linking acute (A) and chronic (B) loneliness to health outcomes. Under normal conditions, acute stress triggers the production of corticotropin releasing hormone (CRH) in the paraventricular nucleus (PVN) of the hypothalamus, which then acts on the anterior pituitary to induce production of adrenocorticotrophic releasing hormone (ACTH). ACTH is released into the bloodstream and acts on the adrenal glands to induce the production of cortisol (the primary stress hormone), among other glucocorticoids. Cortisol then acts on systemic glucocorticoid receptors to produce the stress response. Over time, cortisol accumulates and acts on glucocorticoid receptors in the anterior pituitary, hypothalamus, and hippocampus to negatively regulate the pathway (i.e., stop the stress response) 36. Cortisol also acts on the glucocorticoid receptors present on immune cells to trigger an anti-inflammatory response (i.e., increased production of anti-inflammatory cytokines and decreased production of pro-inflammatory cytokines) 37,38. Under conditions of chronic stress, however, the HPA axis can become dysregulated due to chronically high levels of cortisol. Specifically, glucocorticoid

receptors desensitize and downregulate 39,40, preventing negative regulation of the HPA pathway and allowing for increased production of pro-inflammatory cytokines (IL-6, IL-1, IL-8, TNF- α) 41. Alongside this, stress has been shown to induce inflammation through a number of other pathways including, but not limited to, the upregulation of pro-inflammatory genes and down-regulation of anti-inflammatory genes (i.e., the glucocorticoid receptor gene) and activation of the inflammasome via stress-related neural and endocrine signals 42,43. Pro-inflammatory cytokines such as IL-1 β , TNF- α , and IL-6 have also been shown to positively feedback at the anterior pituitary, hypothalamus, and hippocampus, leading to further activation of the HPA axis 44,45. Iwata et al. conjectured that these pro-inflammatory cytokines disrupt the glucocorticoid receptor-mediated negative feedback on the HPA axis, further increasing cortisol levels within the body 46. Increased levels of pro-inflammatory markers have been associated with an array of adverse health outcomes, including altered brain functioning, impaired cognition, and neurodegeneration 47–49. These cytokines have also been implicated in sickness behaviour, including loss of appetite and fatigue as well as anxiety and dysphoria 50. Alterations in prefrontal cortical function may, in turn, feedback to increase perceptions of social isolation (i.e., loneliness). Stress may also influence poor health behaviours, such as substance use, lack of exercise, smoking etc. It is hypothesized that poor health behaviours arise as a compensatory mechanism to relieve stress and/or as a consequence of limited attentional resources (i.e., individual is too stressed to attend to positive health behaviours); engagement in these activities may feedback to further increase levels of stress and inflammation.

Inclusion Criteria	- Reported an unadjusted association between loneliness and one (or more) health outcome measure(s) - Used either the UCLA Loneliness Scale or the de Jong Gierveld Loneliness Scale (DJGLS), both widely used and validated loneliness scales - Evaluated an outcome derived from clinical interview and/or valid self-report assessment - Used validated measurement scales for health outcome(s) - Cross-sectional, longitudinal, or a randomized controlled trial - All age groups, sexes, ethnicities, and nationalities were included
Exclusion Criteria	- Did not report the main outcome-of-interest (i.e., association between loneliness and health) - Reported single-item measures or did not report a validated health outcome measure - Did not evaluate loneliness (i.e., evaluated a related construct) - Did not measure loneliness using either the UCLA Loneliness Scale or the DJGLS - Evaluated physiological and/or biological markers as outcomes - Did not report original data (e.g., review or protocol registration articles) - Did not report one of our categorical health outcome measures - Did not report the appropriate statistics - Reported the effect of loneliness on health behaviours (e.g., physical activity, healthcare utilization) - Could not access published paper - Duplicate study - Not written in English

Table 1. Inclusion/Exclusion Criteria

Table 2. Pooled effect of loneliness on various health outcomes

	<i>n</i>	Effect size and 95% CI				Heterogeneity			Publication bias (Egger's regression)	
		Point estimate, <i>r</i>	Lower limit	Upper limit	P-value	Q-value	P-value	I ²	Intercept	P-value
Depression	109	-0.497	-0.524	-0.469	<0.001	1649.971	0.000	93.454	-1.902	0.003
Anxiety	18	-0.417	-0.469	-0.362	<0.001	152.107	0.000	88.824	-1.052	0.233
Suicidality	3	-0.516	-0.629	-0.381	<0.001	13.889	0.001	85.600	-9.409	0.002
General Mental Health	21	-0.489	-0.556	-0.415	<0.001	412.924	0.000	95.156	-3.739	0.013
General Health	14	-0.358	-0.470	-0.286	<0.001	702.630	0.000	98.150	3.254	0.149
Well-being	30	-0.476	-0.517	-0.433	<0.001	707.561	0.000	95.901	-2.515	0.027
Physical Health	30	-0.301	-0.348	-0.253	<0.001	389.818	0.000	92.561	-2.031	0.037
Sleep	11	-0.293	-0.335	-0.250	<0.001	21.777	0.016	54.080	-1.190	0.048
Cognition	19	-0.189	-0.262	-0.114	<0.001	414.253	0.000	95.655	-2.481	0.146

Table 3. Results of subgroup analyses (categorical variables; a) and meta-regression analyses (continuous variables; b)

	Age category				Geographical distribution					Stigmatized group			Health condition		
	youth	adults	seniors	<i>p</i>	North America	Europe	Asia	Oceania	<i>p</i>	Yes	No	<i>p</i>	Yes	No	<i>p</i>
Depression	-0.489 (-0.554, -0.418)	-0.469 (-0.525, -0.409)	-0.509 (-0.541, -0.476)	0.475	-0.520 (-0.548, -0.490)	-0.424 (-0.500, -0.342)	-0.491 (-0.564, -0.410)	-0.523 (-0.649, -0.369)	0.135	-0.477 (-0.546, -0.401)	-0.501 (-0.530, -0.471)	0.536	-0.530 (-0.587, -0.469)	-0.490 (-0.520, -0.460)	0.244
	<i>n</i> = 15	<i>n</i> = 18	<i>n</i> = 59		<i>n</i> = 65	<i>n</i> = 22	<i>n</i> = 14	<i>n</i> = 3		<i>n</i> = 22	<i>n</i> = 87		<i>n</i> = 20	<i>n</i> = 89	
Anxiety	-0.407 (-0.533, -0.263)	-0.408 (-0.563, -0.226)	-0.384 (-0.458, -0.304)	0.940	-0.388 (-0.436, -0.338)	-0.447 (-0.592, -0.275)	-0.495 (-0.567, -0.415)	n/a	0.077	-0.461 (-0.503, -0.418)	-0.411 (-0.471, -0.346)	0.184	-0.386 (-0.489, -0.273)	-0.423 (-0.479, -0.363)	0.557
	<i>n</i> = 4	<i>n</i> = 5	<i>n</i> = 4		<i>n</i> = 12	<i>n</i> = 4	<i>n</i> = 2	<i>n</i> = 0		<i>n</i> = 3	<i>n</i> = 15		<i>n</i> = 3	<i>n</i> = 15	
General mental health	-0.522 (-0.745, -0.193)	-0.389 (-0.566, -0.178)	-0.530 (-0.625, -0.420)	0.424	-0.492 (-0.543, -0.436)	-0.425 (-0.555, -0.275)	-0.622 (-0.725, -0.491)	-0.540 (-0.614, -0.457)	0.143	-0.478 (-0.623, -0.302)	-0.492 (-0.568, -0.407)	0.881	-0.474 (-0.600, -0.324)	-0.494 (-0.573, -0.407)	0.799
	<i>n</i> = 2	<i>n</i> = 4	<i>n</i> = 9		<i>n</i> = 9	<i>n</i> = 6	<i>n</i> = 4	<i>n</i> = 1		<i>n</i> = 5	<i>n</i> = 16		<i>n</i> = 6	<i>n</i> = 15	
General health	-0.436 (-0.601, -0.236)	-0.234 (-0.316, -0.148)	-0.287 (-0.332, -0.240)	0.159	-0.312 (-0.368, -0.254)	n/a	-0.425 (-0.648, -0.136)	n/a	0.427	-0.400 (-0.556, -0.217)	-0.354 (-0.471, -0.225)	0.674	-0.450 (-0.554, -0.332)	-0.350 (-0.468, -0.219)	0.242
	<i>n</i> = 6	<i>n</i> = 2	<i>n</i> = 5		<i>n</i> = 9	<i>n</i> = 0	<i>n</i> = 4	<i>n</i> = 0		<i>n</i> = 1	<i>n</i> = 13		<i>n</i> = 1	<i>n</i> = 13	
Well-being	-0.492 (-0.666, -0.268)	-0.480 (-0.613, -0.321)	-0.449 (-0.478, -0.418)	0.852	-0.4100 (-0.429, -0.391)	-0.482 (-0.541, -0.419)	-0.509 (-0.575, -0.436)	-0.500 (-0.579, -0.412)	0.004	-0.471 (-0.542, -0.393)	-0.477 (-0.520, -0.432)	0.893	-0.516 (-0.755, -0.156)	-0.455 (-0.485, -0.424)	0.707
	<i>n</i> = 10	<i>n</i> = 4	<i>n</i> = 15		<i>n</i> = 2	<i>n</i> = 18	<i>n</i> = 8	<i>n</i> = 1		<i>n</i> = 2	<i>n</i> = 28		<i>n</i> = 6	<i>n</i> = 24	
Physical health	n/a	-0.328 (-0.417, -0.233)	-0.297 (-0.353, -0.240)	0.578	-0.299 (-0.366, -0.230)	-0.266 (-0.355, -0.172)	-0.344 (-0.452, -0.226)	n/a	0.584	-0.270 (-0.339, -0.198)	-0.318 (-0.378, -0.256)	0.306	-0.336 (-0.461, -0.198)	-0.290 (-0.345, -0.232)	0.531
	<i>n</i> = 0	<i>n</i> = 8	<i>n</i> = 20		<i>n</i> = 15	<i>n</i> = 8	<i>n</i> = 6	<i>n</i> = 0		<i>n</i> = 11	<i>n</i> = 19		<i>n</i> = 7	<i>n</i> = 22	
Sleep	-0.308 (-0.427, -0.180)	-0.317 (-0.396, -0.233)	-0.266 (-0.355, -0.172)	0.704	-0.347 (-0.394, -0.298)	-0.181 (-0.264, -0.095)	-0.264 (-0.328, -0.197)	-0.290 (-0.463, -0.096)	0.006	-0.360 (-0.481, -0.226)	-0.288 (-0.331, -0.243)	0.304	-0.324 (-0.411, -0.230)	-0.288 (-0.335, -0.240)	0.503
	<i>n</i> = 2	<i>n</i> = 3	<i>n</i> = 4		<i>n</i> = 6	<i>n</i> = 1	<i>n</i> = 3	<i>n</i> = 1		<i>n</i> = 1	<i>n</i> = 10		<i>n</i> = 2	<i>n</i> = 9	
Cognition	n/a	0.006 (-0.197, 0.210)	-0.167 (-0.248, -0.084)	0.125	-0.154 (-0.231, -0.075)	-0.039 (-0.065, -0.013)	-0.495 (-0.617, -0.348)	n/a	<0.001	-0.114 (-0.490, 0.299)	-0.195 (-0.271, -0.117)	0.703	-0.114 (-0.490, 0.299)	-0.195 (-0.271, -0.117)	0.703
	<i>n</i> = 0	<i>n</i> = 2	<i>n</i> = 14		<i>n</i> = 13	<i>n</i> = 3	<i>n</i> = 3	<i>n</i> = 0		<i>n</i> = 2	<i>n</i> = 17		<i>n</i> = 2	<i>n</i> = 17	

Significant regression coefficients are in **bold****a**

	Mean age			Gender distribution (%F)		
	coefficient	CI	p	coefficient	CI	p
Depression	-0.0008	(-0.0030, 0.0014)	0.4781	0.001	(-0.0011, 0.0030)	0.3549
	<i>n</i> = 70			<i>n</i> = 83		
Anxiety	0.0024	(-0.009, 0.0056)	0.1598	0.0001	(-0.0029, 0.0031)	0.9567
	<i>n</i> = 14			<i>n</i> = 16		
General mental health	0.0005	(-0.0059, 0.0068)	0.8838	0.0011	(-0.0071, 0.0092)	0.7957
	<i>n</i> = 13			<i>n</i> = 20		
General health	0.0042	(-0.0022, 0.0106)	0.1987	-0.0027	(-0.0140, 0.0086)	0.6383
	<i>n</i> = 7			<i>n</i> = 13		
Well-being	0.0017	(-0.0012, 0.0046)	0.2504	-0.0011	(-0.0037, 0.0015)	0.4059
	<i>n</i> = 19			<i>n</i> = 30		
Physical health	0.0001	(-0.0052, 0.0054)	0.9699	-0.0016	(-0.0047, 0.0015)	0.3073
	<i>n</i> = 15			<i>n</i> = 29		
Sleep	0.0013	(-0.0014, 0.0040)	0.3431	0.0006	(-0.0018, 0.0030)	0.6276
	<i>n</i> = 8			<i>n</i> = 11		
Cognition	-0.0047	(-0.0118, 0.0025)	0.1998	0.0042	(0.0014, 0.0071)	0.0036
	<i>n</i> = 15			<i>n</i> = 18		

Significant regression coefficients are in **bold**

b

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